

General Sir John Kotelawala Defence University
Faculty of Computing

Ai-Based Smart Campus Decision Support System for Personalized Student Services

Final Report

Module	**CS 22032 (DBA) / CS22023 (CS/SE/COE)**	
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	17 October 2026	

Abstract

Universities generate large amounts of academic information such as semester results, SGPA, repeated courses, attendance, and module requirements. However, students may not always receive timely and personalized support when interpreting this information or planning their studies.

This project presents KDU Academic Intelligence, an AI-based Smart Campus Decision Support System designed to provide personalized academic support through three Artificial Intelligence techniques: a Neural Network, a Genetic Algorithm, and a Rule-Based Expert System. The Neural Network predicts next-semester GPA, the Genetic Algorithm generates personalized study plans by allocating limited weekly study hours, and the Rule-Based Expert System checks academic eligibility using predefined requirements.

The system was implemented using React, TypeScript, Java Spring Boot, Python FastAPI, TensorFlow/Keras, and MySQL. The Neural Network was evaluated using 1,243 held-out records and achieved an MAE of 0.2076, RMSE of 0.2784, and R^2 of 0.6898. A Random Forest baseline achieved an MAE of 0.2207, RMSE of 0.2990, and R^2 of 0.6422. The Genetic Algorithm achieved a final fitness of 29.1932 and 100% correctness in the tested feasibility cases. The components were integrated into a web-based prototype, with the frontend hosted on Vercel and the backend services and database configured to run locally.

The system demonstrates how prediction, optimization, and rule-based reasoning can be combined to support academic planning. The system is intended as a decision-support tool, while final academic decisions remain the responsibility of students and authorized academic staff.

1. Introduction

The increasing availability of digital academic data provides opportunities to use Artificial Intelligence to support university students. Previous research has investigated machine-learning approaches for predicting student academic performance and identifying useful patterns in educational data [1], [2].

Students may need support when estimating their future performance, deciding which modules require additional study time, or checking whether academic requirements have been satisfied. These tasks involve different types of reasoning and therefore cannot always be effectively addressed using a single AI technique.

The proposed system, KDU Academic Intelligence, combines three AI approaches:

1. **Neural Network** - predicts next-semester GPA.
2. **Genetic Algorithm** - optimizes weekly study-hour allocation.
3. **Rule-Based Expert System** - checks academic eligibility.

The system integrates these components into a single web-based application using React, Spring Boot, FastAPI, and MySQL.

The main purpose is not to replace academic advisors or institutional decision-making. Instead, the system provides students with AI-supported information that can assist academic planning and decision-making.

2. Problem Background

2.1 Academic Performance

Students commonly use previous academic results to understand their academic progress. However, manually analysing several semesters of results may not provide an objective estimate of future performance.

Machine-learning techniques have been investigated for predicting academic success using historical student data [1], [2]. Neural Networks can learn relationships between multiple variables and therefore provide a suitable approach for the prediction component.

2.2 Study Planning

Students have a limited number of study hours each week. However, different modules may require different levels of attention depending on academic performance, credit value, academic risk, and repeated-course status.

Genetic Algorithms have been used for personalized learning-path and resource-allocation problems [3], [4]. Therefore, an optimization-based approach can be applied to study-hour allocation.

2.3 Academic Eligibility

Academic activities can require students to satisfy several conditions, including minimum attendance, payment requirements, and prerequisite completion.

Rule-based expert systems can represent such conditions using explicit IF–THEN rules. Previous research has demonstrated the application of rule-based systems to university advising and student support [5], [6].

3. Problem Statement

University students need to make academic decisions using information such as previous performance, current SGPA, repeated courses, attendance, module requirements, and other academic conditions. Manually analysing these factors can be time-consuming and may not provide personalized or timely decision support.

The project therefore addresses three problems:

- **Performance prediction:** estimating next-semester GPA.

- **Study planning:** allocating limited study hours among modules.
- **Eligibility checking:** determining whether academic requirements have been satisfied.

The proposed system addresses these problems using a Neural Network, Genetic Algorithm, and Rule-Based Expert System.

4. Objectives

Main Objective

To develop an AI-based Smart Campus Decision Support System that provides personalized academic support through prediction, study-plan optimization, and academic eligibility checking.

Specific Objectives

1. Develop a Neural Network for next-semester GPA prediction.
2. Develop a Genetic Algorithm for personalized study-hour allocation.
3. Develop a Rule-Based Expert System for academic eligibility.
4. Integrate the three AI components into one web application.
5. Evaluate the AI components using appropriate performance and correctness measures.
6. Provide understandable AI-supported results while maintaining human responsibility for final academic decisions.

5. Selected AI Techniques

The final system uses three AI techniques.

Table 5.1 Selected AI Techniques and Their Applications

Technique	Application
Neural Network	Next-semester GPA prediction
Genetic Algorithm	Personalized study-plan optimization
Rule-Based Expert System	Academic eligibility

5.1 Neural Network

The Neural Network predicts the student's next-semester GPA using:

- Previous SGPA
- Current SGPA
- Repeated Courses
- Current Year

- Current Semester

The target variable is **Next_SGPA**.

The model was implemented using TensorFlow/Keras.

5.2 Genetic Algorithm

The Genetic Algorithm represents a study plan as a chromosome where each gene represents the number of weekly study hours allocated to a module.

For example:

[3, 2, 2, 3, 2, 2, 2, 2, 2]

The algorithm ensures that every target module receives at least one hour and that the total allocation equals the student's available study hours.

The final configuration uses:

Table 5.2 Genetic Algorithm Configuration Parameters

Parameter	**Value**
Population	80
Generations	150
Crossover Rate	0.85
Mutation Rate	0.15
Elitism	2
Tournament Size	4

5.3 Rule-Based Expert System

The rule engine evaluates academic requirements such as:

- Academic profile availability
- Attendance
- Fee payment
- Prerequisite completion

The final output can be:

- **ELIGIBLE**
- **NOT_ELIGIBLE**

- **CONDITIONALLY_ELIGIBLE**

The conditional status is used when prerequisite-related conditions require further completion.

6. Justification of AI Techniques

The three techniques were selected because they solve different types of problems.

Neural Network

A Neural Network can learn relationships between multiple academic variables and predict a continuous value such as GPA. Machine-learning approaches have been investigated for academic performance prediction [1], [2].

Genetic Algorithm

Study planning is a constrained optimization problem. A Genetic Algorithm can search through possible allocations while satisfying constraints. Genetic Algorithms have been investigated for personalized learning paths and academic resource allocation [3], [4].

Rule-Based Expert System

Eligibility requirements are explicit conditions that can be naturally represented using IF–THEN rules. Rule-based systems are also suitable when explanations are required [5], [6].

Therefore, the project demonstrates three different AI concepts:

Learning → Optimization → Reasoning

7. System Design

7.1 Architecture

The final system consists of four main layers:

- React + TypeScript frontend
- Java Spring Boot backend
- Python FastAPI AI services
- MySQL database

The main data flow is:

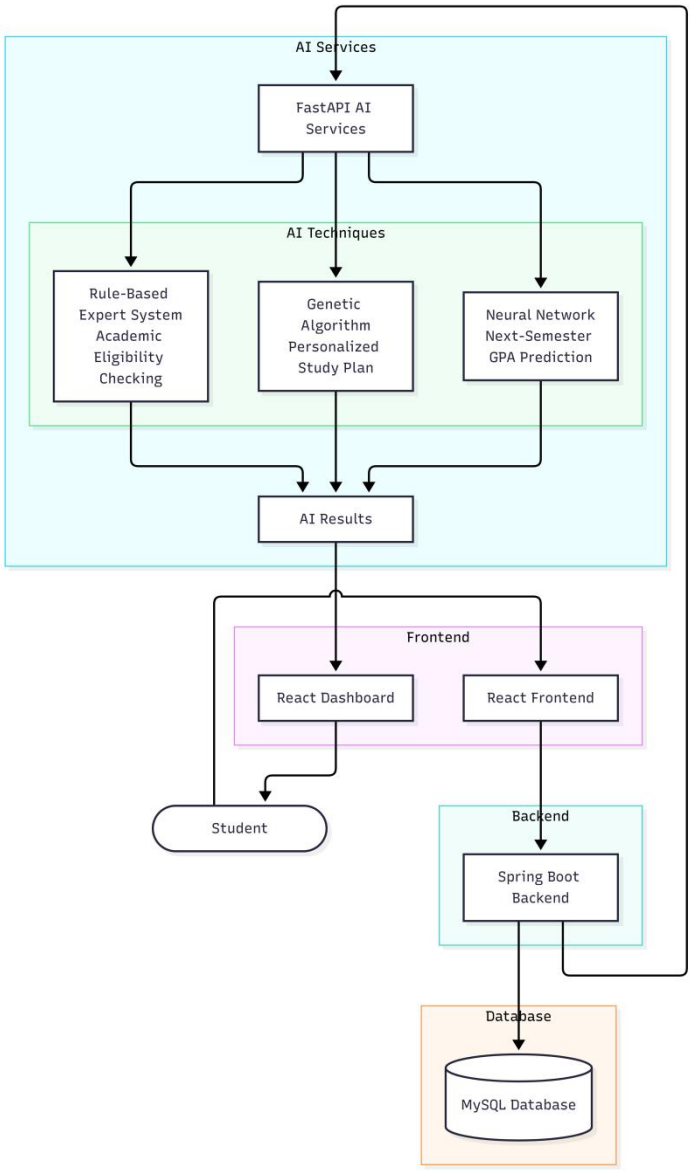


Figure 7.1 Final System Architecture

7.2 Frontend

The frontend was developed using React, TypeScript, and Vite.

The main functions include:

- Student dashboard
- GPA prediction
- Study-plan generation
- Eligibility checking
- AI result presentation

7.3 Backend and AI Services

The Java Spring Boot backend manages application requests, database operations, and communication between the frontend and backend services. Python FastAPI exposes the Neural Network-based GPA prediction and Genetic Algorithm-based study-plan generation functionalities through REST APIs. The Rule-Based Expert System is implemented within the Spring Boot backend and evaluates academic eligibility using predefined rules.

This architecture separates machine-learning inference and optimization from the main application logic while integrating the three AI techniques into one system.

7.4 Database

MySQL stores structured academic information.

- **Broader academic dataset:** approximately 3,000 students, 108,083 result records, 220 modules, and 345 degree-module relationships.
- **Prediction dataset:** 6,129 records representing 2,026 unique students.

8. Implementation Details

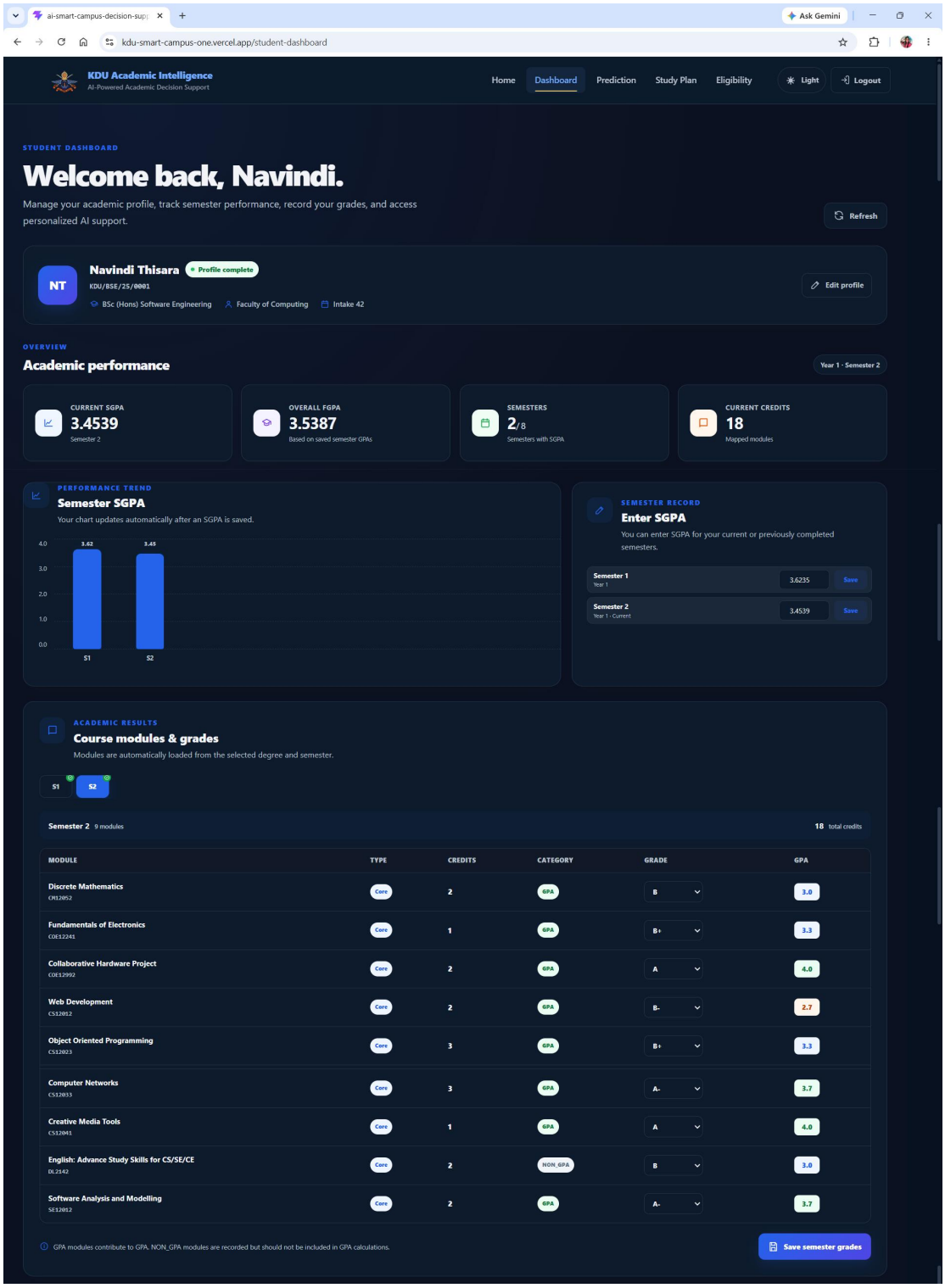
8.1 Technology Stack

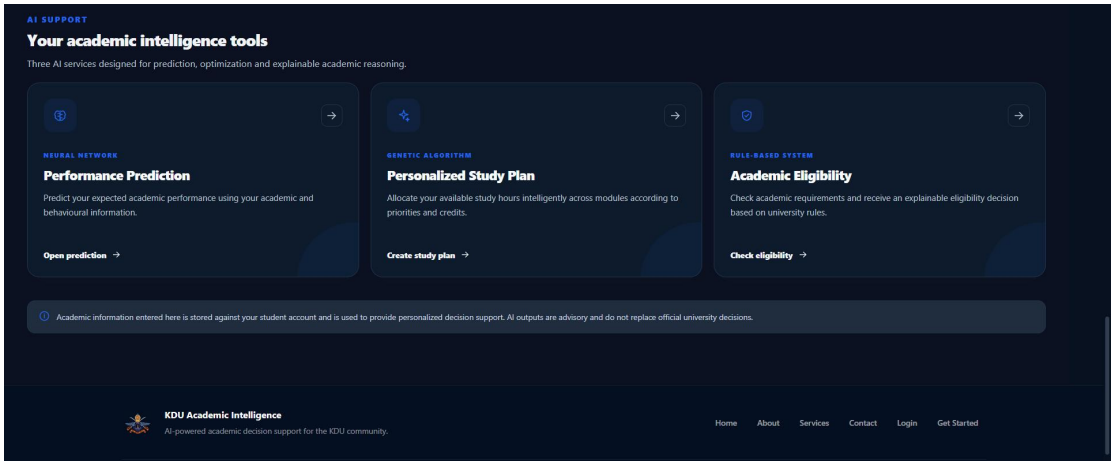
Table 8.1 Technology Stack of the Proposed System

Component	Technology
Frontend	React, TypeScript, Vite
Backend	Java, Spring Boot
AI Services	Python, FastAPI
Machine Learning	TensorFlow/Keras
Data Processing	Pandas, NumPy
Evaluation	Scikit-learn
Database	MySQL
Version Control	Git/GitHub
Deployment	Vercel (frontend); local deployment for Spring Boot, FastAPI and MySQL

8.2 User Interface and System Integration

The final web application provides a student dashboard through which users can access the AI-based academic services. The dashboard presents student information, GPA prediction, study-plan generation, and eligibility checking.





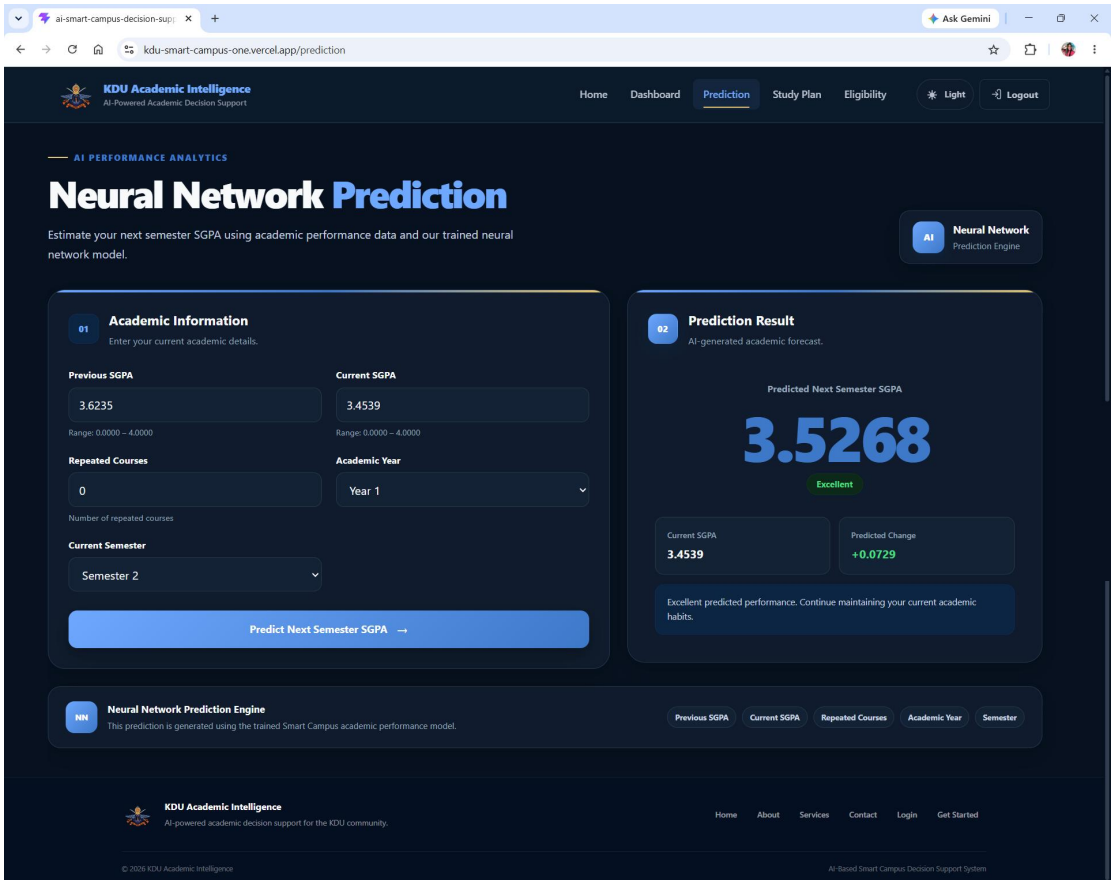


Figure 8.2.1 Final Student Dashboard and GPA Prediction

8.3 Neural Network Implementation

The final prediction dataset contains:

- 6,129 records
- 2,026 unique students

The dataset contains the following main attributes:

Student_ID
Degree_ID
Intake
Gender
Current_Year
Current_Semester
Previous_SGPA
Current_SGPA
Repeated_Courses
Next_SGPA

Dataset validation produced 10/10 successful validation checks, including missing-value, GPA-range, duplicate, semester-consistency, and target-leakage checks.

The model uses:

- Adam optimizer
- Mean Squared Error loss
- Batch size 32
- Maximum 200 epochs
- Early stopping
- StandardScaler

The final model was evaluated using a held-out student-level test set.

8.4 Genetic Algorithm Implementation

The GA calculates module priorities using:

- Module credits
- Module type
- Historical performance
- Academic risk
- Repeated-course information

The fitness function considers the difference between generated study hours and target hours, utilization of available hours, and excessive concentration.

The simplified fitness structure is:

Fitness =
30

− Weighted Allocation Error
− Balance Penalty
+ Utilization Bonus

The algorithm uses tournament selection, one-point crossover, mutation, elitism, and a repair mechanism to maintain feasible solutions.

8.5 Rule-Based Implementation

The system applies explicit rules.

Example:

IF attendance >= 80%
AND fee_status = PAID
AND prerequisites = COMPLETED
THEN eligibility = ELIGIBLE

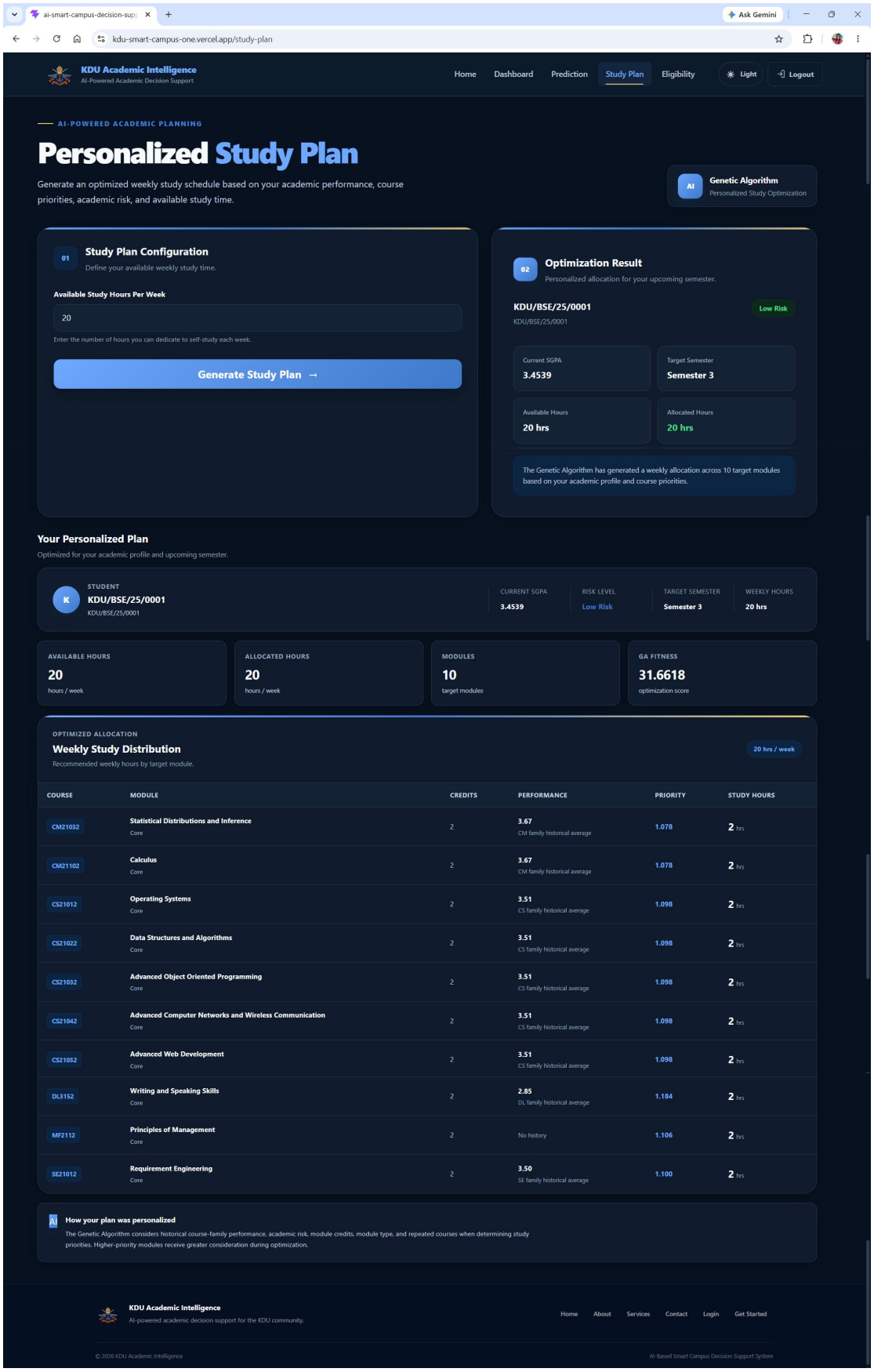
When required prerequisites are incomplete but other mandatory requirements are satisfied, the system can return:

CONDITIONALLY_ELIGIBLE

The system also provides an explanation of the relevant conditions.

8.6 Personalized Study Plan and Eligibility Interface

The system presents the outputs of the Genetic Algorithm and Rule-Based Expert System through the student interface. The study-plan interface displays the recommended weekly allocation of study hours across modules, while the eligibility interface presents the eligibility status and relevant conditions.


ai-smart-campus-decision-sup
Ask Gemini
kdu-smart-campus-one.vercel.app/study-plan
KDU Academic Intelligence
AI-Powered Academic Decision Support
Home
Dashboard
Prediction
Study Plan
Eligibility
Light
Logout
AI-POWERED ACADEMIC PLANNING
Personalized Study Plan
Generate an optimized weekly study schedule based on your academic performance, course priorities, academic risk, and available study time.
AI
Genetic Algorithm
Personalized Study Optimization
01
Study Plan Configuration
Define your available weekly study time.
Available Study Hours Per Week
20
Enter the number of hours you can dedicate to self-study each week.
Generate Study Plan →
02
Optimization Result
Personalized allocation for your upcoming semester.
KDU/BSE/25/0001
KDU/BSE/25/0001
Low Risk
Current SGPA
3.4539
Target Semester
Semester 3
Available Hours
20 hrs
Allocated Hours
20 hrs
The Genetic Algorithm has generated a weekly allocation across 10 target modules based on your academic profile and course priorities.
Your Personalized Plan
Optimized for your academic profile and upcoming semester.
K
STUDENT
KDU/BSE/25/0001
KDU/BSE/25/0001
CURRENT SGPA
3.4539
RISK LEVEL
Low Risk
TARGET SEMESTER
Semester 3
WEEKLY HOURS
20 hrs
AVAILABLE HOURS
20
hours / week
ALLOCATED HOURS
20
hours / week
MODULES
10
target modules
GA FITNESS
31.6618
optimization score
OPTIMIZED ALLOCATION
Weekly Study Distribution
Recommended weekly hours by target module.
20 hrs / week
COURSE
MODULE
CREDITS
PERFORMANCE
PRIORITY
STUDY HOURS
CM21032
Statistical Distributions and Inference
Core
2
3.67
CM family historical average
1.078
2 hrs
CM21102
Calculus
Core
2
3.67
CM family historical average
1.078
2 hrs
CS21012
Operating Systems
Core
2
3.51
CS family historical average
1.098
2 hrs
CS21022
Data Structures and Algorithms
Core
2
3.51
CS family historical average
1.098
2 hrs
CS21032
Advanced Object Oriented Programming
Core
2
3.51
CS family historical average
1.098
2 hrs
CS21042
Advanced Computer Networks and Wireless Communication
Core
2
3.51
CS family historical average
1.098
2 hrs
CS21052
Advanced Web Development
Core
2
3.51
CS family historical average
1.098
2 hrs
DL3152
Writing and Speaking Skills
Core
2
2.85
DL family historical average
1.184
2 hrs
MF2112
Principles of Management
Core
2
No history
1.106
2 hrs
SE21012
Requirement Engineering
Core
2
3.50
SE family historical average
1.100
2 hrs
AI
How your plan was personalized
The Genetic Algorithm considers historical course-family performance, academic risk, module credits, module type, and repeated courses when determining study priorities. Higher-priority modules receive greater consideration during optimization.
KDU Academic Intelligence
AI-powered academic decision support for the KDU community.
Home
About
Services
Contact
Login
Get Started
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AI-Based Smart Campus Decision Support System

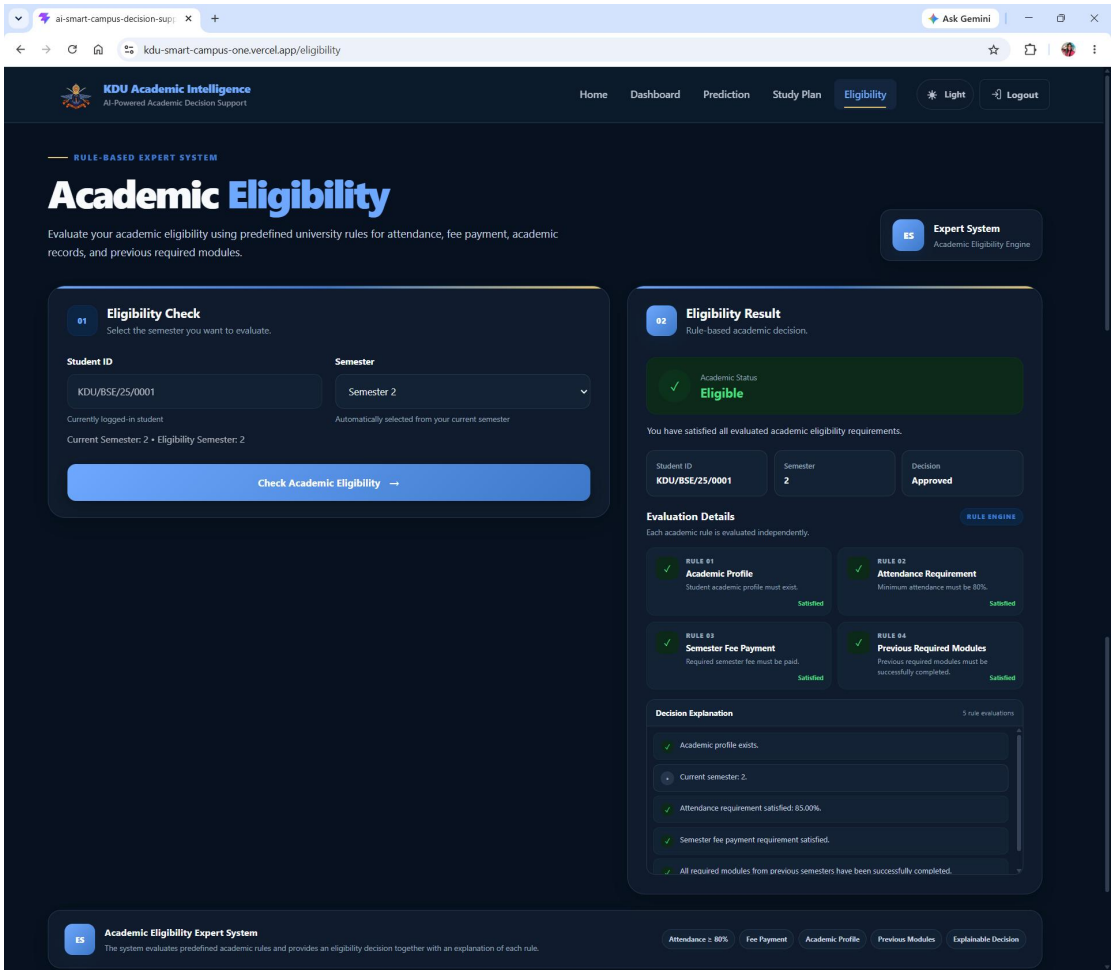


Figure 8.6.1 Personalized Study Plan and Academic Eligibility Results

9. Testing and Results

9.1 Neural Network Results

The Neural Network was evaluated using 1,243 held-out test records.

Table 9.1 Neural Network and Random Forest Performance Comparison

Metric	Neural Network	Random Forest
MAE	0.2076	0.2207
RMSE	0.2784	0.2990
R^2	0.6898	0.6422

The Neural Network achieved lower MAE and RMSE and higher R^2 than the Random Forest baseline for the evaluated dataset.

Prediction Error

Table 9.2 Neural Network Prediction Error Distribution

Error Range	Percentage
Within ±0.10	32.50%
Within ±0.20	60.98%
Within ±0.30	77.15%
Within ±0.50	91.87%

The model performed better in the higher-GPA ranges. The MAE was 0.1620 for the 3.50–4.00 range but 0.6534 for observations below 2.00.

This difference is important because the lower-GPA range contains substantially fewer observations.

9.2 Genetic Algorithm Results

Convergence

Table 9.3 Genetic Algorithm Convergence Results

Measure	Result
Initial Best Fitness	27.52938
Final Best Fitness	29.19321
Improvement	6.04%
Generations	150
Population	80

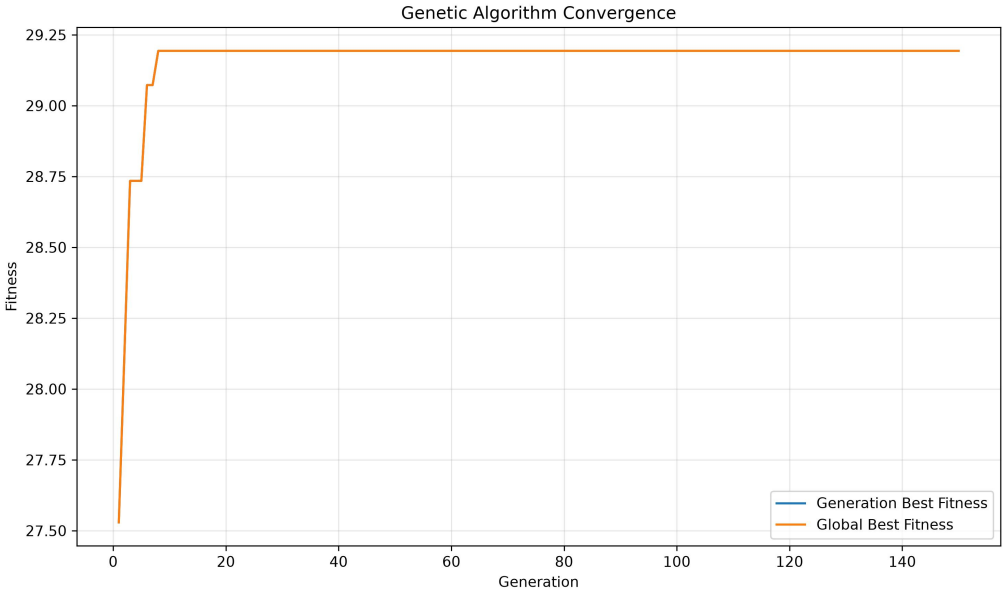


Figure 9.1 GA Convergence

Multi-Seed Testing

Thirty random seeds were tested.

Table 9.4 Genetic Algorithm Multi-Seed Testing Results

Measure	Result
Mean Fitness	29.19321
Standard Deviation	Approximately 0
Minimum Fitness	29.19321
Maximum Fitness	29.19321
Mean Absolute Hour Error	0.266777

All tested seeds produced the same final fitness for the evaluated problem instance.

Baseline Comparison

Table 9.5 Comparison of Study-Plan Allocation Approaches

Approach	Fitness	MAE
Equal Allocation	27.501875	0.457074
Priority Allocation	29.193207	0.266777
Genetic Algorithm	29.193207	0.266777

The GA improved the project-defined fitness by 1.6913 points compared with equal allocation.

The GA and priority allocation produced the same result in the evaluated scenario.

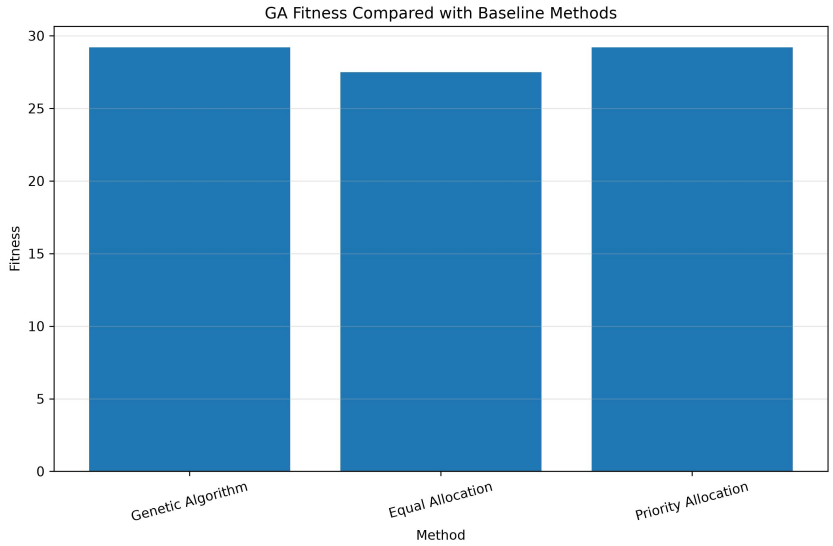


Figure 9.2 GA Baseline Comparison

Feasibility Testing

The feasibility test achieved:

100% correctness

The algorithm correctly rejected allocations where available hours were less than the number of target modules and accepted feasible allocations.

9.3 Rule-Based Testing

Representative test cases include:

Table 9.6 Rule-Based Expert System Test Cases and Expected Results

Condition	Expected Result
All requirements satisfied	ELIGIBLE
Attendance below 80%	NOT_ELIGIBLE
Fee unpaid	NOT_ELIGIBLE
Prerequisite missing	CONDITIONALLY_ELIGIBLE
Academic profile unavailable	NOT_ELIGIBLE

The rule engine provides deterministic outputs based on its defined rules and can explain the conditions leading to the result.

9.4 Integration Testing

The system components were tested in the local development environment to verify communication between the frontend, Spring Boot backend, MySQL database, and FastAPI AI services. The tested functionalities included GPA prediction, personalized study-plan generation, eligibility checking, and the display of AI-generated results.

The frontend is hosted on Vercel, while the Spring Boot backend, AI APIs, and MySQL database currently run locally because the Railway trial has expired. Therefore, the hosted frontend should not be considered a fully operational end-to-end deployment unless it is connected to an accessible backend and database.

10. Limitations & Discussion

10.1 Dataset

The prediction dataset is a project/synthetic dataset and is not an official university-wide operational dataset. Therefore, the reported performance should not be interpreted as guaranteed real-world performance.

10.2 Prediction Model

The final model uses five academic features. Variables such as attendance, assignment marks, study hours, and personal circumstances are not included in the final GPA prediction model.

The model also performs less accurately for the relatively small low-GPA group.

10.3 Genetic Algorithm

The GA depends on the project-defined fitness function and priority factors. A different fitness function or additional academic constraints could produce different study plans.

The algorithm also cannot guarantee that a student will follow a generated plan or that following the plan will improve academic results.

10.4 Rule-Based System

The rule engine only represents the academic conditions implemented in the knowledge base. Actual university regulations may contain additional requirements.

10.5 Generalization

Further evaluation using larger, diverse, and independent real-world datasets would be required before using the system for operational academic decision-making.

11. Ethical and Social Considerations

11.1 Privacy

Student academic records should only be accessible to authorized users. Data used for model development should also be appropriately protected.

11.2 Bias

The prediction dataset is not evenly distributed across all GPA ranges. The model therefore requires careful interpretation, particularly for groups with fewer observations.

11.3 Accuracy

AI predictions are estimates and contain errors. The reported MAE of 0.2076 demonstrates that the model should not be treated as a source of guaranteed future GPA.

11.4 Explainability

The rule-based component provides explanations for eligibility decisions. The study-planning component uses identifiable academic factors to determine module priorities.

11.5 Human Oversight

The system is a decision-support tool. It should not automatically fail students, apply penalties, approve academic requests, or replace authorized academic decision-makers.

12. Conclusion

This project developed KDU Academic Intelligence, an AI-based Smart Campus Decision Support System combining three AI approaches: Neural Network prediction, Genetic Algorithm optimization, and Rule-Based reasoning.

The Neural Network achieved an MAE of 0.2076, RMSE of 0.2784, and R^2 of 0.6898 on the evaluated test set. The Genetic Algorithm successfully generated feasible study plans and achieved stable results across the tested random seeds. The Rule-Based Expert System provided transparent eligibility outcomes based on predefined academic conditions.

The three components were integrated using React, TypeScript, Spring Boot, FastAPI, TensorFlow/Keras, and MySQL and deployed as a web-based prototype.

The project demonstrates that different AI techniques can be combined to address different aspects of academic decision support. However, the system remains a prototype and should not replace official university procedures or human academic judgement.

Future development could include larger real-world datasets, additional prediction features, more comprehensive university rules, improved optimization objectives, continuous model evaluation, and integration with official university information systems.

13. References

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[2] J. H. Guanin-Fajardo, J. Guaña-Moya, and J. Casillas, "Predicting Academic Success of College Students Using Machine Learning Techniques," *Data*, vol. 9, no. 4, p. 60, Apr. 2024, doi: 10.3390/data9040060.

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[5] M. Patankar, "A Rule-Based Expert System Approach to Academic Advising," *Innovations in Education and Training International*, vol. 35, no. 1, pp. 49–58, Feb. 1998, doi: 10.1080/1355800980350107.

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Presentation Slides & Video: https://drive.google.com/drive/folders/1eKgjvR4x0-CkLR3tMRte0RyJBkkAQ6Xv?usp=drive_link

Github Repository: https://github.com/Navindi-Thisara/AI-Smart-Campus-Decision-Support-System.git

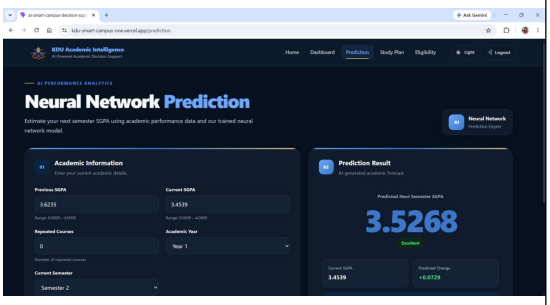
ESSENTIALS OF ARTIFICIAL INTELLIGENCE

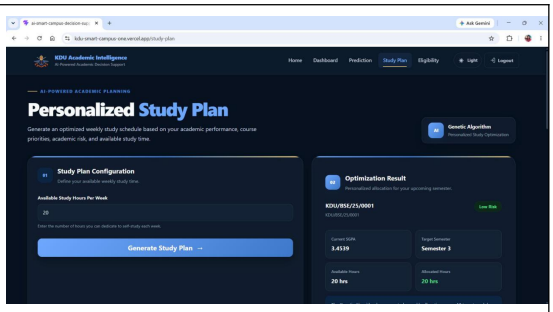
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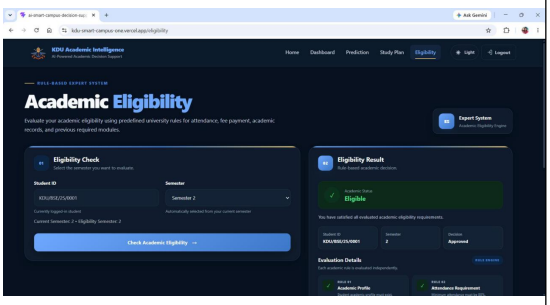
Student Name	JLN Thisara	**Registration No.**	D/BSE/25/0005
Degree / Intake	BSc (Hons) in Software Engineering / 42	**Group No.**	20
Project Title	AI-Based Smart Campus Decision Support System for Personalized Student Services		

Technical Contributions:

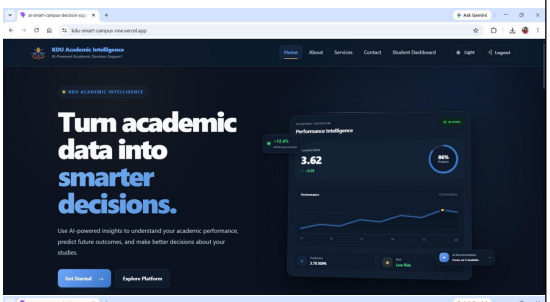
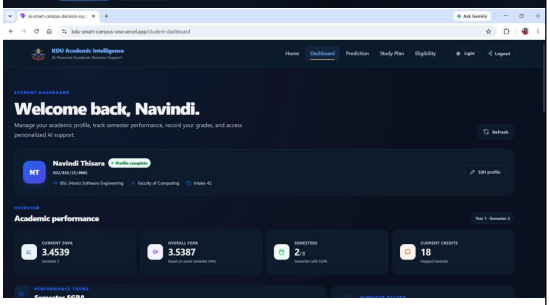
Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Dataset Preparation and Validation	Prepared and validated the academic dataset for GPA prediction. Performed data cleaning and checked missing values, GPA ranges, duplicates, semester consistency, and target leakage.	Python, Pandas, NumPy	Dataset validation source code scripts/final_ml_dataset_validation.py Dataset cleaning source code scripts/validate_and_clean_next_sgpa.py Validation report data/ml/final_ml_validation_report.txt Cleaned dataset data/ml/next_semester_gpa_dataset_clean.csv
2. Neural Network Development	Developed a Neural Network to predict students' next-semester GPA using Previous SGPA, Current SGPA, Repeated Courses, Current Year, and Current Semester.	Python, TensorFlow/Keras, Neural Network	Model training and development scripts/train_next_sgpa_model.py Prediction implementation scripts/predict_next_sgpa.py Saved Neural Network models/next_sgpa_neural_network.keras

			Model input features models/next_sgpa_features.txt 
3. Neural Network Evaluation	Evaluated the model using a held-out test dataset and compared its performance with a Random Forest baseline using MAE, RMSE, and R^2.	Scikit-learn, TensorFlow/Keras	Evaluation implementation scripts/final_next_sgpa_evaluation.py Final evaluation report data/ml/final_next_sgpa_evaluation_report.txt Model comparison CSV data/ml/next_sgpa_model_comparison.csv Detailed evaluation CSV data/ml/next_sgpa_detailed_evaluation.csv
4. Genetic Algorithm Development	Implemented a Genetic Algorithm to generate personalized weekly study plans by allocating available study hours among target modules.	Python, Genetic Algorithm	GA implementation scripts/study_plan_ga.py Study-plan API app/study_plan_api.py Generated study-plan interface src/pages/StudyPlan.tsx

			
5. Genetic Algorithm Optimization and Testing	Implemented the fitness function, tournament selection, crossover, mutation, elitism, and repair mechanisms. Evaluated convergence, feasibility, and baseline performance.	Python, Genetic Algorithm	GA evaluation and experiments scripts/evaluate_ga.py Final GA evaluation report data/ga_evaluation/GA_Final_Evaluation_Report.txt Baseline comparison data/ga_evaluation/ga_baseline_comparison.csv Feasibility tests data/ga_evaluation/ga_feasibility.csv Convergence results data/ga_evaluation/ga_convergence.csv
6. Rule-Based Expert System	Developed an eligibility engine using predefined rules to evaluate academic profile availability, attendance requirements, fee payment status, and prerequisite completion.	Java, Spring Boot, Rule-Based Expert System, IF–THEN Rules	Eligibility business rules backend/src/main/java/com/smartcampus/backend/service/EligibilityService.java Eligibility status definitions backend/src/main/java/com/smartcampus/backend/service/EligibilityStatus.java Eligibility API controller backend/src/main/java/com/smartcampus/backend/controller/EligibilityController.java

			Frontend eligibility interface src/pages/Eligibility.tsx 
7. AI Service Development	Developed FastAPI services to expose the Neural Network-based GPA prediction and Genetic Algorithm-based study-plan generation functionalities through REST APIs for integration with the main application.	Python, FastAPI, REST APIs	GPA prediction API app/next_sgpa_api.py Study-plan generation API app/study_plan_api.py GPA prediction logic scripts/predict_next_sgpa.py Dependencies requirements.txt
8. Backend and Database Integration	Integrated the AI services with the Java Spring Boot backend and connected the application to the MySQL database for academic data operations.	Java, Spring Boot, MySQL	Main Spring Boot application BackendApplication.java Student API controller StudentController.java GPA controller StudentSemesterGpaController.java Student results repository StudentResultRepository.java Database configuration

			application.properties
9. Frontend Integration	Integrated the AI functionalities into the student interface so users could access GPA predictions, personalized study plans, and academic eligibility results.	React, TypeScript, Vite, CSS	Main React application src/App.tsx Student dashboard src/pages/StudentDashboard.tsx GPA prediction interface src/pages/Prediction.tsx Study-plan interface src/pages/StudyPlan.tsx Eligibility interface src/pages/Eligibility.tsx
10. System Testing and Deployment Verification	Performed functional and integration testing of the implemented system components and verified the main student-service functionalities. Tested the Spring Boot backend and AI API services in the local development environment and verified the frontend deployment on Vercel. Following the expiration of the Railway free trial, the backend services and MySQL database were configured to run locally.	Functional Testing, Integration Testing, Spring Boot Testing, React, FastAPI, MySQL, Vercel, Local Deployment.	Existing Spring Boot test BackendApplicationTests.java Backend build configuration backend/pom.xml Vercel configuration vercel.json Project instructions README.md Hosted frontend https://kdu-smart-campus-one.vercel.app/ **Deployment status:** Frontend hosted on Vercel; Spring Boot backend, AI APIs,

			and MySQL database running locally.  
11. Technical Documentation, Project Proposal, Progress Report and Final Report	Contributed to the technical documentation and project reports covering system implementation, AI model evaluation, testing, limitations, ethical considerations, and results, according to my actual writing and development contributions.	Microsoft Word, Git/GitHub, Technical Documentation	Project documentation README.md Neural Network evaluation report data/ml/final_next_sgpa_evaluation_report.txt GA evaluation report data/ga_evaluation/GA_Final_Evaluation_Report.txt

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: Navindi **Date:** 16.10.2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	GRT Perera	**Registration No.**	D/BIS/24/0017
Degree / Intake	BIS / 41	**Group No.**	20
Project Title	AI-Based Smart Campus Decision Support System for Personalized Student Services		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Rule-Based Expert System Testing	Performed functional testing of the academic eligibility-checking component.	Rule-Based Expert System, IF–THEN Rules, Functional Testing.	Verified eligibility-checking functionality
2. Eligibility Test-Case Execution	Tested attendance below 80%, unpaid fees, missing prerequisites, unavailable academic profiles, and satisfied requirements.	Rule-Based Logic, Manual Test Cases	Five eligibility test scenarios
3. Eligibility Output Verification	Checked eligibility statuses against expected results.	Rule Engine, Expected vs. Actual Result Comparison	Verified ELIGIBLE, NOT_ELIGIBLE, and CONDITIONALLY_ELIGIBLE outputs.
4. Integration-Result Verification	Checked eligibility results and	React, TypeScript, FastAPI,	Eligibility interface & integration testing

	conditions displayed through the application interface.	Integration Testing	
5. Testing Documentation	Documented test cases, expected outcomes, observed results, and verification activities.	Test-Case Documentation, Functional Testing	Rule-Based Test Cases and final report documentation.

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** 16-10-2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	USPP Shavindi	**Registration No.**	D/DBA/25/0026
Degree / Intake	BSc (Hons) in Data Science and Business Analytics / 42	**Group No.**	**20**
Project Title	AI-Based Smart Campus Decision Support System for Personalized Student Services		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Neural Network Functional Testing	Performed functional testing of the Neural Network-based GPA prediction component to verify that the prediction functionality operated as expected for the prepared test cases.	Functional testing; Neural Network (GPA prediction service)	Functional test results recorded in the testing documentation
GPA Prediction Output Verification	Examined the prediction outputs returned by the system and checked that the results were displayed correctly through the application's interface.	Output verification; application interface	Recorded output verification results (expected vs. actual predictions)

Prediction Test-Case Preparation	Prepared and executed test cases for the GPA prediction functionality to examine its behaviour with different academic input values.	Test-case design and execution; varied academic input values	Test case table with input values, expected results and execution results
Result Verification	Assisted in verifying the displayed prediction results and identified unexpected behaviour during the testing process.	Result verification; defect/behaviour observation	Written record of observed results and unexpected behaviour
Testing Documentation	Documented the prediction testing activities, observed results, and relevant verification outcomes for inclusion in the project report.	Report documentation	GPA prediction testing section of the project report

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 16/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	YRSS Kumara	**Registration No.**	D/BIT/24/0040
Degree / Intake	BSc (Hons) in Information Technology / 41	**Group No.**	20
Project Title	AI-Based Smart Campus Decision Support System for Personalized Student Services		

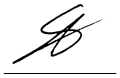
Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Genetic Algorithm testing	Conducted functional testing on the genetic algorithm-based study plan generation module to check whether the algorithm gave study-hour allocations for the targeted modules.	Functional testing; Genetic Algorithm (study plan generation)	Results of functional test documented on study plan testing document
Output Verification	Verifying the study plans to check whether the study-hour allocations per week were displayed properly through the application interface	Output verification; application interface	Results of verification on study plan output

Execution of Feasibility Test Cases	Conducted tests on the validity and invalidity of the algorithms for study hour's allocation cases including cases whereby the number of hours available was less than that of the targeted modules.	Feasibility testing; valid and invalid study-hour allocation cases	Test case table of inputs, expected feasibility outputs and execution results.
Baseline Comparison Verification	Helped in verifying the results of the equal allocation, priority allocation and genetic algorithm-based allocation which included the fitness value and mean absolute hour error.	Baseline comparison; fitness value; mean absolute hour error	Verifying baseline comparison results as recorded for the project
Testing Documentation	Documented the testing processes for the study plans and the results as well as the verification process for the project.	Report documentation	Part of the project report on genetic algorithm study plan testing

Declaration

I hereby declare that the contributions made above are a true reflection of the individual contribution of mine in this particular group assignment.

Student Signature  **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.